

Panel MS-AR(1) with random-effects intercepts, 1970Q1 onward

1 Model

Joint panel Markov-switching AR(1) around a *linear* trend, with three regimes:

$$y_{it} = a_i + g t + z_{it}, \tag{1}$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it}) z_{it} + \sigma(s_{it}) \varepsilon_{it}, \tag{2}$$

$$s_{i,t+1} \sim \Pi(\cdot \mid s_{it}), \tag{3}$$

$$a_i \sim \mathcal{N}(\alpha, \omega^2) \quad (\text{random effects}). \tag{4}$$

The trend is linear in calendar time t (g per year, common across countries). In the pooled specification $a_i \equiv a$. In the random-effects specification the a_i are i.i.d. Normal draws; α and ω are estimated by maximum likelihood, integrating each country's Hamilton-filter likelihood by 11-point Gauss–Hermite quadrature. Cycles are plotted using the posterior mean of a_i . Latent paths s_{it} are country-specific. The regime dated t governs the transition from z_t to z_{t+1} . σ and ρ switch with the regime. The median μ is pinned at 0. $|\rho| < 0.995$. Standard errors (in parentheses) are delta-method from a numerical Hessian on shared parameters. π is the ergodic distribution of Π . $E[z]$ is the long-run mean of the cycle implied by μ , ρ , and Π .

2 Common intercept and linear trend

SOE sample from 1970Q1: log real GDP per worker. 62 countries, 6924 observations, 1970-Q1–2024-Q3. Fitted countries: 62. Observations: 6924. Log-likelihood: 16652.31. Ergodic mean of the cycle $E[z] = 0.3934$ (0.1970). Common intercept $a = -3.8544$ (0.1934). Common slope $g = 0.0124$ (0.0009).

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.6834 (0.1942)	0.0000 —	1.3119 (0.2051)
σ	0.0109 (0.0002)	0.1086 (0.0050)	0.0242 (0.0004)
ρ	0.9950 (0.0000)	0.9950 (0.0000)	0.9950 (0.0000)
$\Pi_0.$	0.9706 (0.0036)	0.0198 (0.0035)	0.0096 (0.0031)
$\Pi_1.$	0.0714 (0.0192)	0.7053 (0.0386)	0.2233 (0.0376)
$\Pi_2.$	0.0149 (0.0032)	0.0206 (0.0038)	0.9646 (0.0044)
π	0.4181 (0.0366)	0.0642 (0.0085)	0.5177 (0.0360)

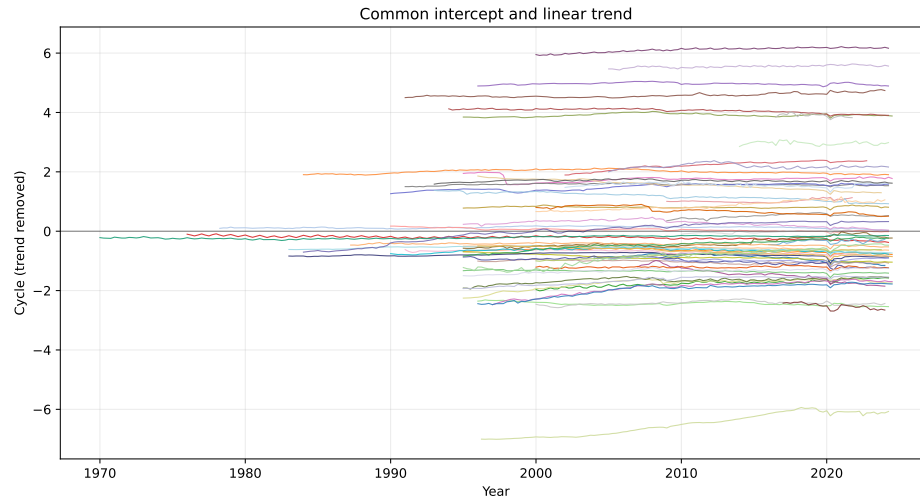


Figure 1: Country cycles after removing the linear trend.

3 Random-effects intercepts, common linear trend

SOE sample from 1970Q1: log real GDP per worker. 62 countries, 6924 observations, 1970-Q1–2024-Q3. Fitted countries: 62. Observations: 6924. Log-likelihood: 17673.12. Ergodic mean of the cycle $E[z] = 0.2136$ (0.0460). Random intercepts $a_i \sim N(\alpha, \omega^2)$ with $\alpha = -3.9761$ (0.0693), $\omega = 1.1868$ (0.0133). Posterior-mean a_i : mean -3.514, min -8.631, max 2.181. Common slope $g = 0.0084$ (0.0007).

Table 2: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.5933 (0.2441)	0.0000 —	0.5486 (0.0620)
σ	0.0680 (0.0031)	0.0220 (0.0006)	0.0097 (0.0002)
ρ	0.9950 (0.0000)	0.9950 (0.0000)	0.9950 (0.0000)
Π_0	0.7820 (0.0317)	0.1448 (0.0299)	0.0732 (0.0177)
Π_1	0.0201 (0.0046)	0.9581 (0.0060)	0.0218 (0.0045)
Π_2	0.0202 (0.0035)	0.0124 (0.0039)	0.9673 (0.0038)
π	0.0847 (0.0121)	0.4352 (0.0337)	0.4801 (0.0346)

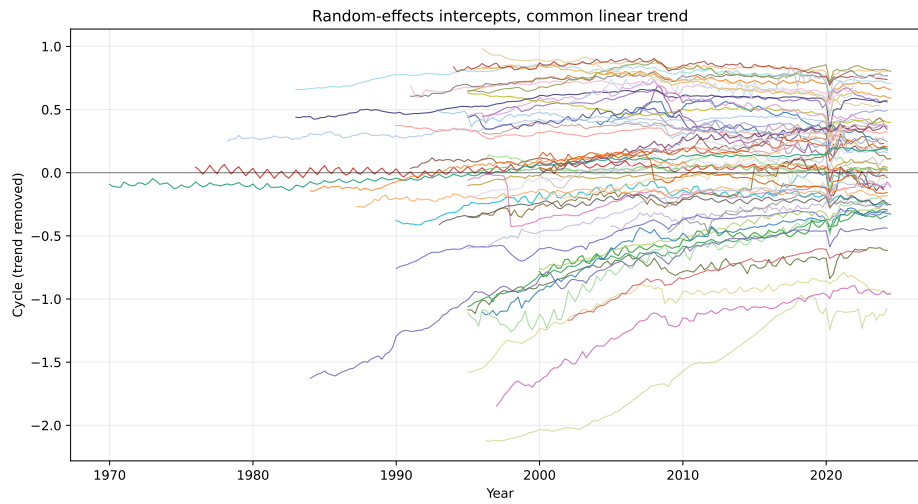


Figure 2: Country cycles after removing the linear trend.